

Akshera Paladhi

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EDUCATION

University of California Santa Cruz, Santa Cruz, CA

Sept 2024 – Expected June 2028

B.S. Electrical Engineering, Minor in Astrophysics

GPA: 3.58/4.0 (Dean's Honor List)

- **Relevant Coursework:** Electronic Circuits, Introduction to Electricity Industry, Computer Systems and C Programming, Computer Systems and Assembly Language, Scientific Computing, Introduction to Physics I: Mechanics, Introduction to Electricity and Magnetism, Waves, Optics, Thermodynamics, Statics, Dynamics, Linear Algebra, Differential Equations

SKILLS

- **Software & Programming:** Strong computer proficiency in C, C++, Python, Altium, MATLAB, STM32, PIC32, Arduino IDE, Linux (Bash/Shell), Git, Siemens NX, KiCad, SQL, Nvidia Isaac, Mujoco, SolidWorks, Onshape, STM32Cube
- **Engineering & Technical Tools:** Oscilloscopes, Multimeters, Microcontrollers (RP2040, ESP32), FPGA, VHDL, Verification & Validation (V&V) testing, Root Cause Analysis, EHS Standards, Precision Soldering
- **Certifications:** Certificate of Completion NPWEE Summer 2025 L'SPACE (August 2025 – Present)

EXPERIENCE

Citrus Drone Initiative for Drone Education and Research Internship

Santa Cruz, CA

Drone Researcher

July 2026 - Present

- Designed airframes and mounts in Onshape and SolidWorks, optimizing CAD models for rapid 3D printing (Bambu Lab, PrusaSlicer)
- Assembled, wired, and soldered flight hardware and power distribution systems for custom multicopter testbeds
- Earned FAA Part 107 Remote Pilot Certification; preparing for upcoming group flight and engineering research

UCSC Institute of Particle Physics (SCIPP)

Santa Cruz, CA

Undergraduate Researcher

June 2025 – Present

- Applying circuit analysis to simulate and continuously improve the robustness of the high-speed front-end electronics using LTspice for the Diamond Submerged Wireless Imaging Monitor (SWIM) project
- Designing a signal-shaping circuit by applying schematic design and debugging skills to stabilize ultra-short detector pulses for proton beam diagnostics at Los Alamos National Lab (LANL), ensuring reliable performance
- Developing C firmware on an Arduino Nano RP2040 Connect for real-time data acquisition to monitor particle beam profiles and resolve hardware-software integration issues
- Leveraged Python scripting in a Linux to automate real-time testing of Thorlabs cameras, evaluating sensor response and data integrity across various wavelengths

UC Santa Cruz Quantum Computing

Santa Cruz, CA

Vice President/Researcher

August 2025 – Present

- Leading a 10+ member research group in the study of Position-Based Quantum Cryptography (PBQC) utilizing strong interpersonal and organizational skills, applying analytical models to develop secure authentication frameworks
- Collaborating with 3+ UC campuses and the Quantum Coalition to expand shared research resources

PROJECTS

Interactive Household Robotics (Lamp & Alarm Clock)

July 2026 – Present

- Designing custom multi-DOF robot CAD in Onshape, optimizing enclosures and joints for DFM/DFA and 3D printing
- Conducting static FEA, vibration analysis, and kinematic calculations to validate dynamic stability and acoustic performance
- Engineering circular 2-layer PCBs in KiCad with ESP32-S3 microcontrollers, I2S audio, and power regulation for real-time actuation

Autonomous Drone Warehouse Surveyor

June 2026 – Present

- Designing custom multicopter CAD geometry in SolidWorks with integrated forward and downward camera mounts
- Setting up physics, mass properties, and collision primitives in NVIDIA Isaac Sim via USD/URDF conversion
- Scripting autonomous waypoint navigation, thruster controls, and camera feeds using Omniverse ActionGraph

Air Mouse (ESP32 + MPU-6050)

Dec 2025 – Jan 2026

- Developing a wireless motion-controlled robotics interface using an ESP32 microcontroller and MPU-6050 IMU
- Implemented sensor fusion and custom filtering algorithms in C to translate real-time inertial data into precise navigation via Bluetooth HID integration
- Designed a custom PCB in KiCad and executing precision SMD soldering to deliver production-ready hardware